

COVID-19 Mortality Trend Analysis

Indhira Vadivel

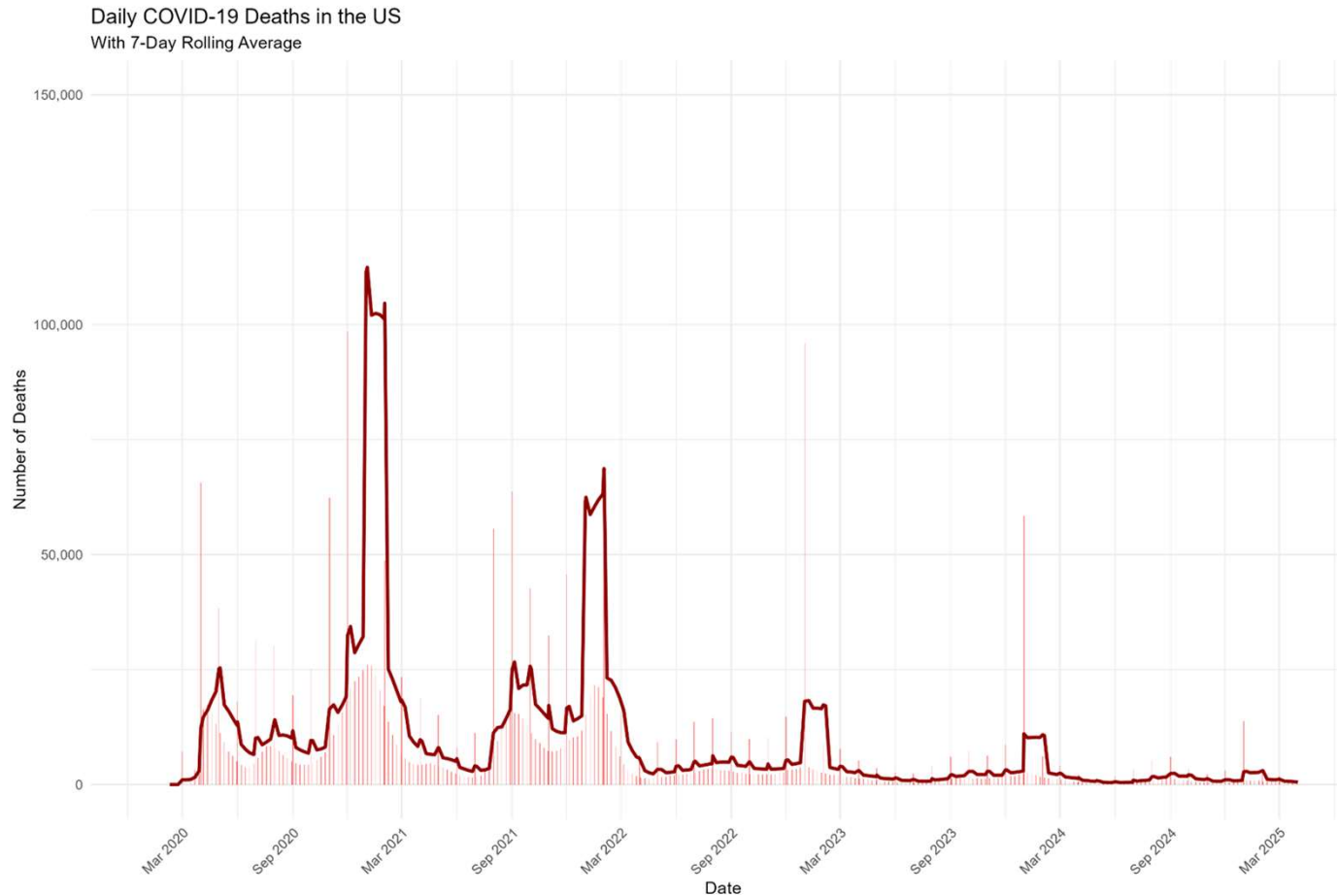
Data source:

Covid_19 deaths & Pneumonia Deaths (2013-2019): <https://data.cdc.gov/>

Population(2020-2024): <https://api.census.gov/data>

Visualizing the death trend (2020-2025) and classifying the periods into waves

COVID-19 pandemic in the US (2020-2025)

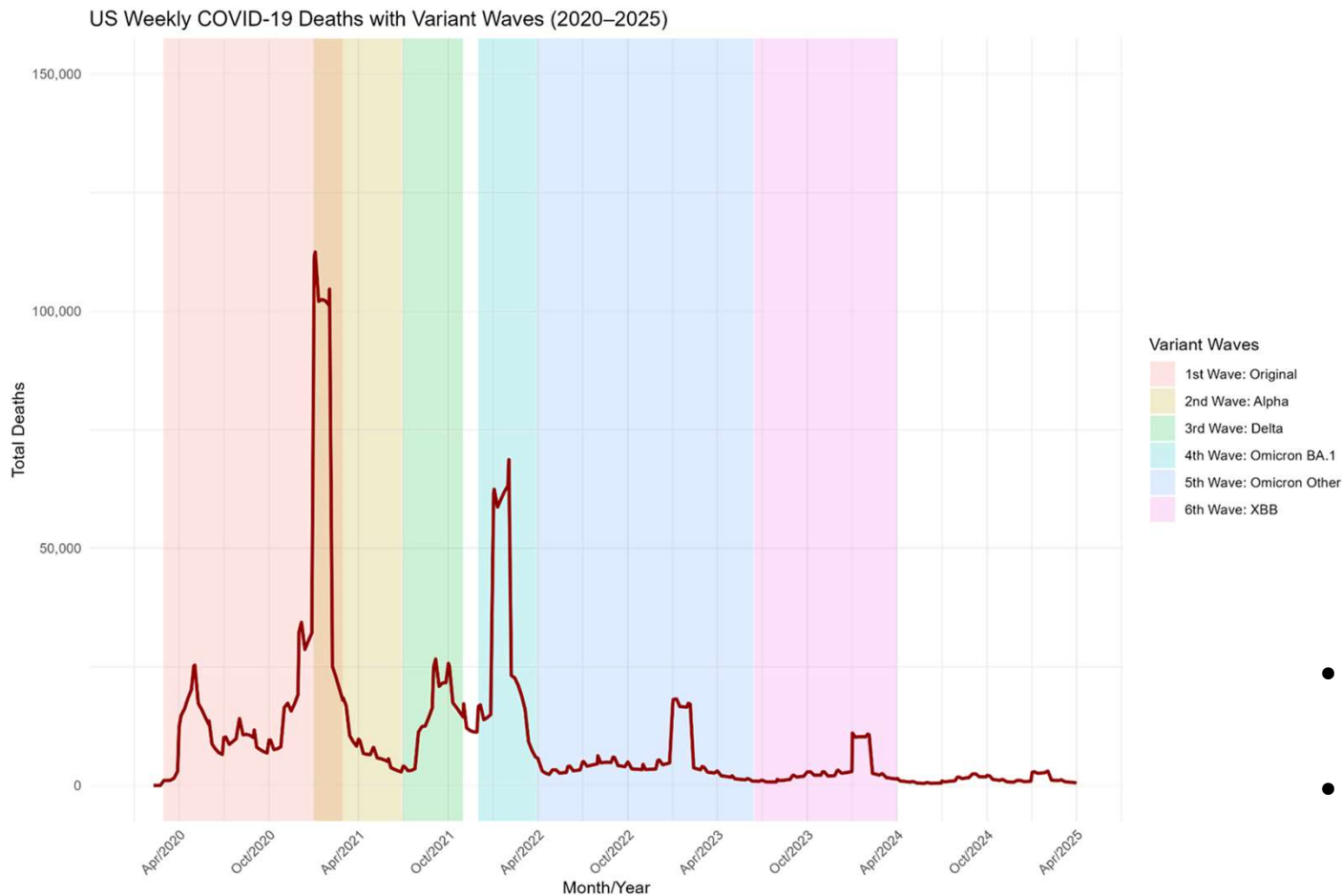


This line graph shows the **trend** in COVID-19 deaths in the U.S. over the period from 2020 to 2025.

Highlights:

- A large wave of deaths began in early 2020
- Rolling average highlights a sharp increase and peak, showing the early impact.
- Starting around 2022, there is a **notable decrease** in both daily deaths and their average. Suggests improved control: due to **vaccinations**, **treatment improvements**, and **public health measures**.
- The death counts become relatively low and stable toward the end of the timeline. Very small fluctuations indicate **low transmission and fatality levels**.

Dividing the pandemic period, January 2020 to April 2025 into waves



Periods of peaks and troughs	Variant	Wave
Mar 2020 – Feb. 2021	Original Wuhan strain + early mutations	1st Wave
Jan 2021 – June 2021	Alpha	2nd Wave
July 2021 – Oct 2021	Delta	3rd Wave
Dec 2021 – Mar 2022	Omicron BA.1	4th Wave
Apr 2022 – Early 2023	Omicron BA.2 → BA.5 → XBB	5th Wave
Mid-2023 – Early 2024	Omicron XBB lineage	6th Wave

- Visualize peaks and patterns aligned to corresponding waves at that particular time period.
- Several significant peaks follow, likely aligning with the **Alpha**, **Delta** and **Omicron** variants.
- Each wave shows a **steep rise** and gradual decline in deaths

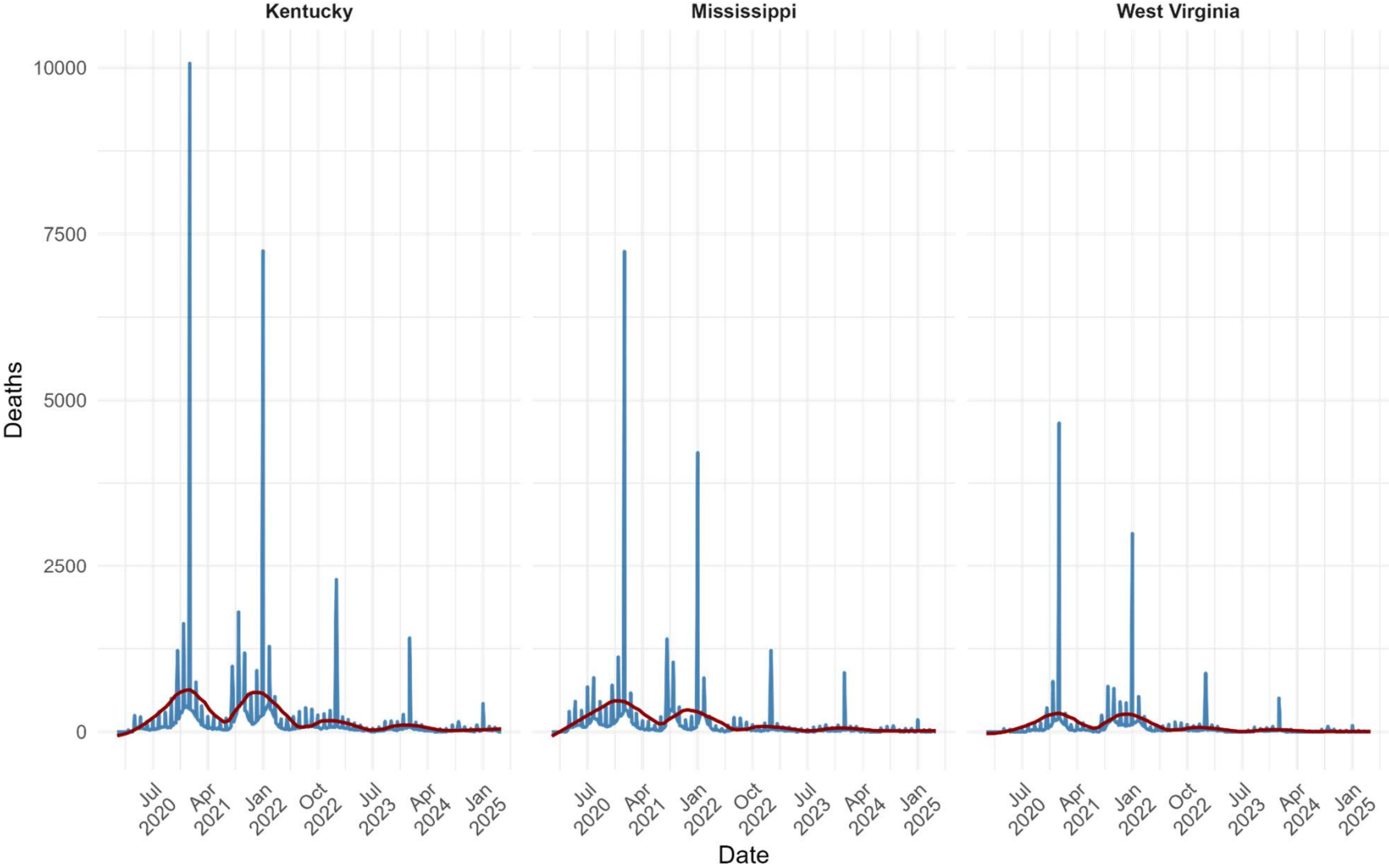
Heatmap showing intensity of deaths by state(Quarterly)



- Many states show dark bands in early 2020 and late 2021, aligning with major COVID surges (e.g., original outbreak, Delta, and Omicron waves).
- Most states experienced highest death tolls during certain quarters, often corresponding to winter surges (e.g., Q4 2020, Q1 2021))
- States like Texas, Florida, California, and Pennsylvania consistently show brighter tiles, signaling larger absolute death counts, possibly due to population size and state-specific outbreaks.
- Other states like Vermont, Hawaii, and Alaska remain lighter, indicating fewer deaths overall.
- By 2023 and beyond, many states show lighter colors, suggesting a reduction in COVID-19 deaths, potentially due to widespread vaccination, prior immunity, and improved treatment

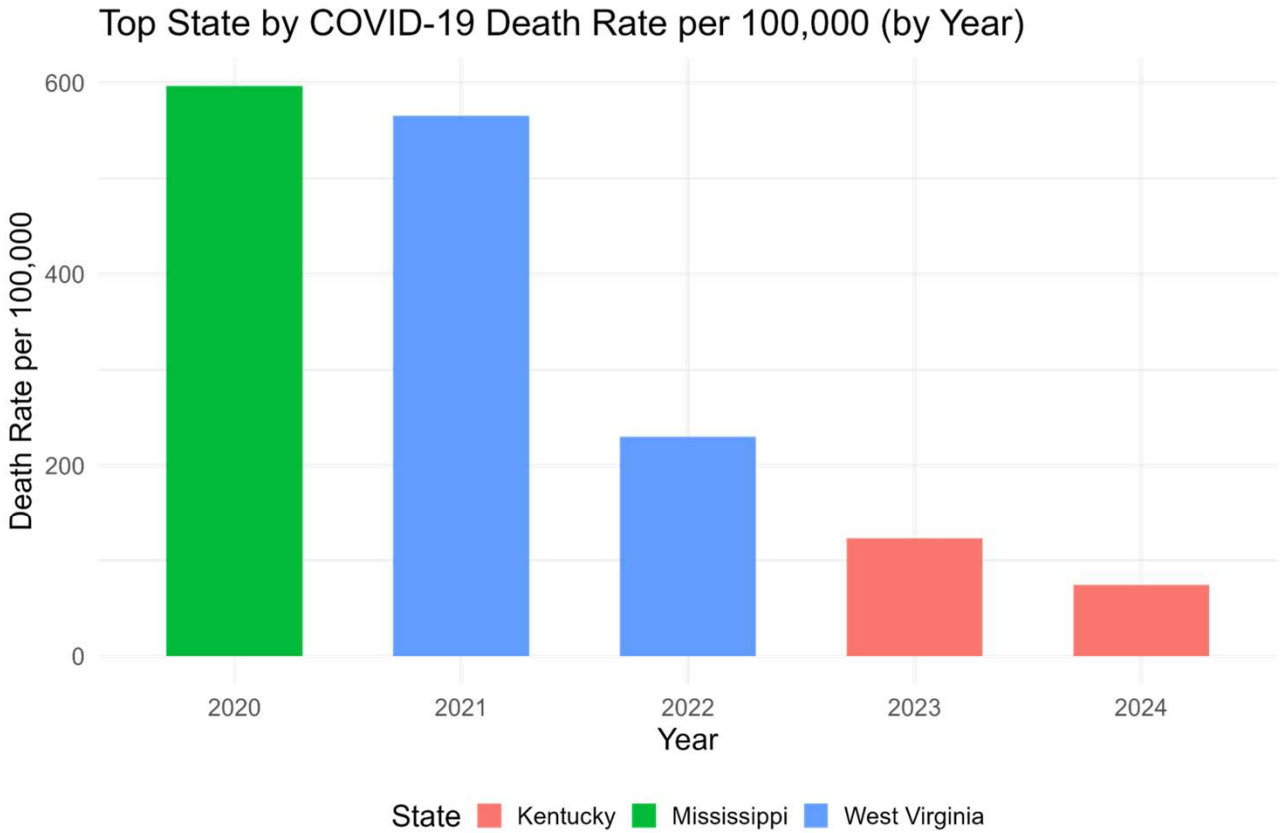
Quantifying the Impact (Covid-19 Deaths) across states and regions

COVID-19 Deaths Over Time – Faceted by States



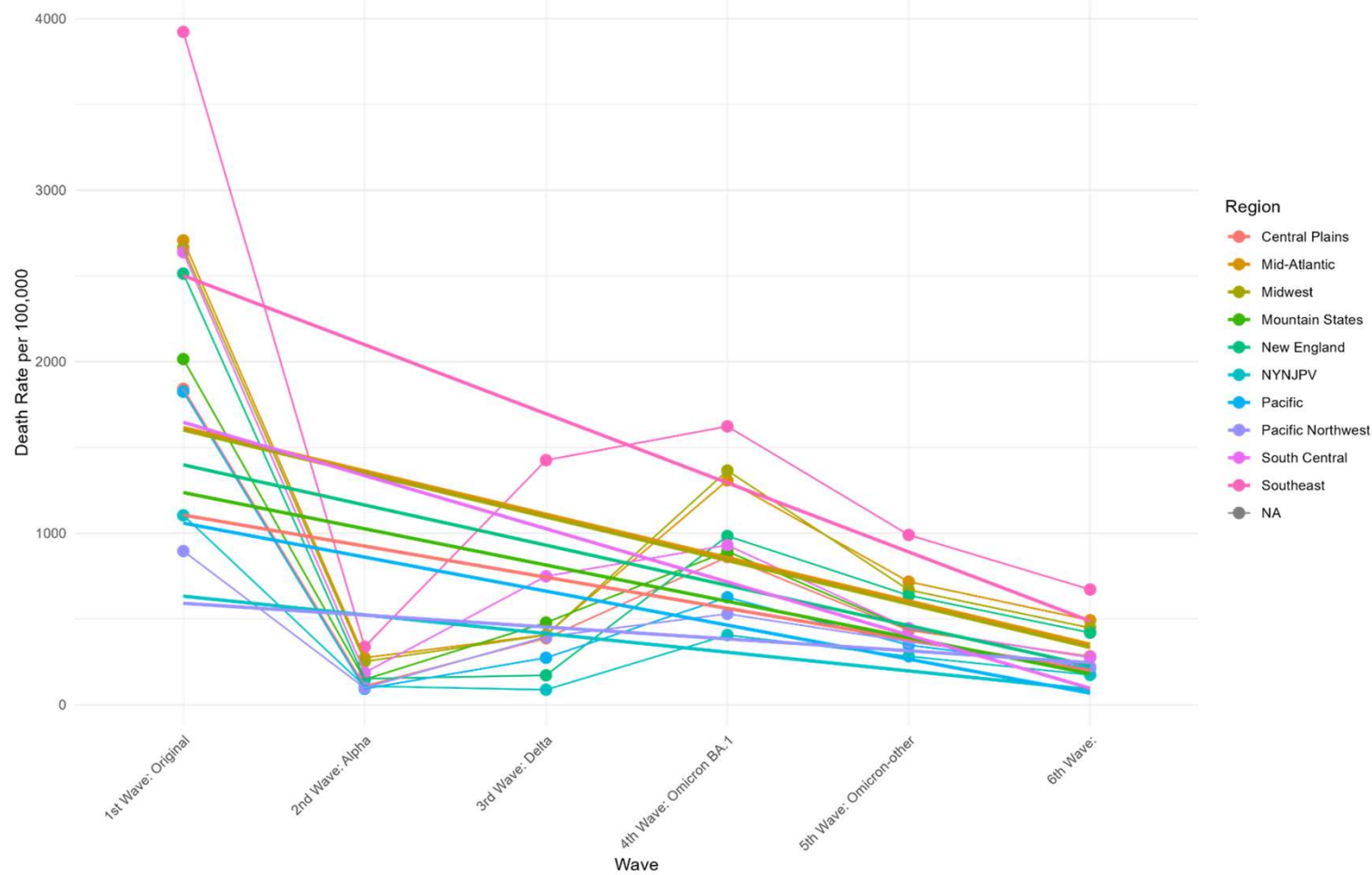
Summary - States with highest death rate (by year)

Year	State	Covid_Deaths	Population	Death_rate_per_100k
2020	Mississippi	17641	2956870	596.61061
2021	West Virginia	10084	1782959	565.57666
2022	West Virginia	4084	1774122	230.19837
2023	Kentucky	5597	4550595	122.99491
2024	Kentucky	3406	4588372	74.23112



Analyzing the trends across different waves

Linear Trend of Death Rates Across COVID-19 Waves by Region



Plot components:

- Each line shows how a **regional** death rate changed from wave to wave
- To avoid clutter, lines are not labeled, but group trends are visible

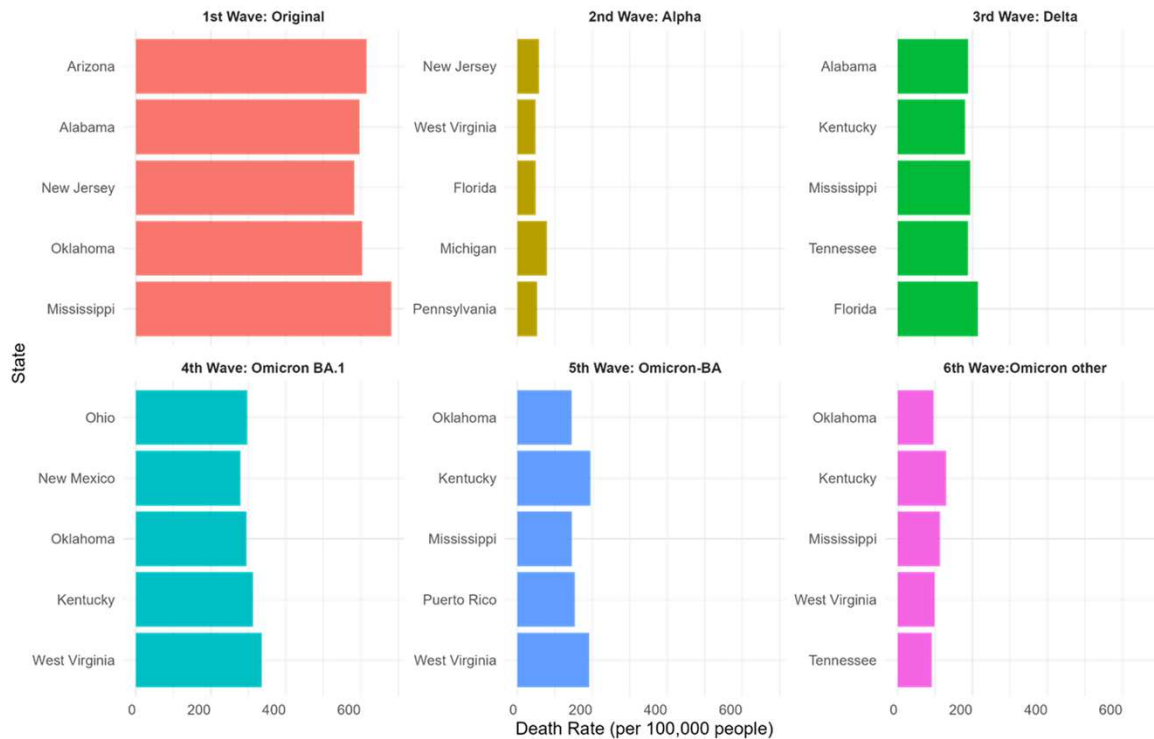
The **slope** of a line indicates whether a region's death rate is increasing, decreasing, or staying flat over successive COVID-19 waves.

- **Downward trend:** Indicates improvement over time (declining death rates).
- **Upward trend:** Worsening situation or late impact.
- **Flat line:** Relatively stable death rate across waves.

Top 5 States by COVID-19 Death Rate for Each Variant Wave

This plot gives a **clear snapshot of the top 5 hardest-hit states** (by death rate per 100,000 people) during **each major COVID-19 variant wave**

Top 5 States by COVID-19 Death Rate for Each Variant Wave



Top States by COVID-19 Death Rate per 100,000 population

State	Wave	Death Rate
Mississippi	1st Wave: Original	681.0
Michigan	2nd Wave: Alpha	78.8
Florida	3rd Wave: Delta	214.2
West Virginia	4th Wave: Omicron BA.1	336.4
Kentucky	5th Wave: Omicron-BA	194.9
Kentucky	6th Wave: Omicron other	129.6

Main Summary Insights

1. The data shows that most U.S. states experienced **peak COVID-19 deaths** during major surges, particularly **in early 2020 and late 2021**, aligning with waves like the original outbreak, Delta, and Omicron.
2. The most affected states based on highest death rate by year are **Mississippi(2020), West Virginia(2021 & 2022), Kentucky(2023,2024)**.
3. States in **South Central and Southeast US** consistently had **highest death rate** compared to other regions across successive Covid Waves. Some of the regions had linear decreasing trend whereas some have mixed trends.
4. States like **Mississippi and Kentucky** consistently had **highest death rates** and showed up in top 5 hardest hit states requiring intervention and improvement in public health measures.
5. **Virulency decreased across waves** based on hospitalizations (severity) and fatality (case fatality rate)
6. Percent Excess mortality was **highest during the initial and Delta waves** of COVID-19 in the U.S., with Western regions experiencing relatively higher rates compared to Eastern regions.
7. Reported **COVID-19 deaths do not explain all excess mortality**, and possible misclassification with pneumonia also fails to fully account for the observed excess deaths. **Pneumonia is likely a secondary effect of Covid-19.**